

Lightweight Hybrid Neural Architecture for Real-Time Deepfake Screening and Content Integrity Preservation

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Abstract—With the help of sophisticated artificial intelligence (AI), the so-called deepfakes, both photos and videos, are becoming more and more real and pose a threat to the credibility, privacy, and integrity of content of the population. Although very advanced, current deepfake detectors are typically overly biased towards either spatial or temporal delays, are typically based on intensive computing architectures, and lack interpretability, which limits their usefulness in real-world applications, resilience and ability to apply to a wide range of lives. In order to fill the abovementioned gaps, this paper suggests a hybrid neural network to preserve content integrity and to screen real-time deepfakes. The proposed architecture will be a hybrid of Long Short-Term Memory (LSTM) sub-model that should be able to detect temporal distortions, e.g., irregular facial movement and abnormal motion pattern, and a MobileNet backbone that can help extract spatial features in a more efficient way. When trained on a large and heterogeneous dataset of authentic and manipulated samples, MobileNetLSTM framework is based on the state-of-the-art accuracy, which is 97% with its precision, recall, and F1-scores of 93, 94, and 93, respectively. Due to the lightweight design, the proposed model can be deployed on a resource-constrained platform with low-latency inference, which can be used in digital forensics, social media theaters, and content checks. Also, explainable AI (XAI) approaches based on Grad-CAM heatmaps are included to identify anomalous regions on the faces that impact model outputs, which increases clarity and trust in the users. All in all, this work offers a high-performance, scalable, and interpretable way to enhance the authenticity of digital media, and future research should focus on multimodal deepfake analysis, continuous learning to keep up with new forms of manipulation and be more resilient to the new generative models.

Keywords—Hybrid Neural Architecture, Real-Time Screening, Content Integrity Preservation, Temporal Feature Analysis, Spatial Feature Extraction, Media Authentication

I. INTRODUCTION

The fast-moving evolution of AI based generative technologies has emerged revolutionarily in terms of synthetic media creation. Techniques are now making it possible to generate incredibly realistic images and videos, and often indistinguishable from the real thing [1]. While these innovations have advantageous uses in entertainment, media and creative industries, they have at the same time posed significant risks to society [2]. The large accessibility of these tools has led to the creation and dissemination of malicious deepfake content, especially on social media platforms, and this poses great threats for issues related to privacy and content integrity and threatens public trust. Recent reports indicate the exponential expansion of the use of deepfake videos for blackmail, fraud, or the creation of pornographic videos of public figures, as well as deepfakes used to manipulate regular people through fake videos of a relative [3]. Alarming, research shows that a significant portion of the public is still unaware of the consumer manipulations that are happening: 71% of people trust videos potentially made up. Reliable detection systems are therefore of critical importance.

To thwart against these threats a range of deepfake detection techniques have been developed such as CNNs, ViT, Linformer variants, and hybrid models [4]. These approaches mostly aim to capture spatial inconsistent aspects and time inconsistent aspects of manipulated media, e.g. irregular facial movements, unnatural facial expressions, lighting inconsistency, etc. While such models have shown great levels of accuracy under controlled conditions, they often face a number of limitations: Many models are computationally expensive, and can only be used over short-timeframes; others lack robust generalization, and therefore performance declines upon exposure to unknown techniques of deepfake generation. Moreover, the present methods focus commonly on static images or audio-only data, leaving some

room for discovering video-based deepfakes, where both the spatial and temporal analysis play an important role [5].

Overcoming such limitations, this work proposes a lightweight hybrid neural architecture consisted with the MobileNet backbone for spatial feature extraction with the LSTM module to enjoy the temporal dynamics such as irregular facial movements, as well as subtle inconsistencies in the temporal dimension. The MobileNet and LSTM consists of a deep neural network optimized for real-time inheritance on computational-hobbled downside devices, as these are not regulated by calculation resources thus keeping the high degree of detection accuracy and minimum of calculation resources. This makes it extremely suitable to be deployed in digital forensics, social media monitoring, news verification pipelines, law enforcement applications.

A. Work Contributions

a) Development of Hybrid Architecture: A novel fusion of MobileNet and LSTM architectures for the combined analysis of spatial temporal features by deepfake videos.

b) Increased Robustness and Generalization: Improved ability to take in multiple sources of data and a wide range of deepfake creation techniques.

c) Computational Efficiency: Lower processing costs that allow computation to be in real time without being more degrading to detection accuracy.

d) Explainable AI Integration: The strategy of incorporating Grad-CAM heatmaps to show regions that affect predictions for a better understanding and improved trust by users and to improve interpretability.

II. LITERATURE REVIEW

Geeta Rani et al., (2025) the convolutional neural networks, transformer models and the Linformer-based architectures have all been studied in the previous deepfake detection research and they all have advantages in the spatial feature extraction and attention-based analysis. However, as the technology to create deepfakes improves and often requires large computational resources, most methods show reduced accuracy. According to some of the latest research, hybrid architectures, when Linformer's lower attention complexity meets the efficiency of transformers, betoken the robustness, generalization, and real-time performance of deepfake detection tasks [6].

Siddiqui Muhammad Yasir et al., (2025), in an attempt to deal with the growing threats that are manipulated media have posed, most previous studies that tried to detect deepfakes focused on deep learning approaches and visual feature analysis. Keyframing, machine learning classifiers and texture descriptions such as HOG, LBP, KAZE have been proven to be able to increase accuracy yet reduce computation costs. Recent studies deem the development of effective and portable deepfake detection solutions necessary by showing that feature-level fusion and inefficient models can achieve high performance on conventional datasets [7].

Recently, we are seeing that the literature is giving particular attention to the rapid developments in deepfake generation and limited approaches to secure the authenticity of digital media. Studies from 2018-2024 focus on detection techniques covering video, image and audio analysis with

focuses mostly on deep learning, computer vision, and other advanced machine learning techniques. These works show in total the increasing trend towards autonomous, robust and multi-modal deepfake detection systems to fight ever more sophisticated manipulations [8].

Georgios Petmezas et al., (2025) suggested that the deepfake detection has become more dependent on deep learning approaches using spatial plus temporal analysis. Recent studies combine CNNs, LSTMs, and Transformers, sometimes combined with facial features models to enhance accuracy and efficiency for a variety of quality of the video and types of manipulation. These works emphasize the tendency toward hybrid architectures which deal with the local and global temporal information for strong and real-time detection [9].

Muhammad Javid et al., (2024) have proposed the recent Deepfake detection research delves into the exploration of hybrid deep learning models that integrate lightweight CNN with a deep neural network models to capture the subtle facial manipulations and complex visual features. Approaches such as MesoNet and ResNet hybrid sometimes enhanced with eye movement analysis have shown a high accuracy and real-time potential for video forensics and verification of content integrity [10].

Hanan Saleh Alhaji et al., (2024) have proposed the recent studies in the detection of deepfakes focus on hybrid methods that use optimization-based feature extraction approaches such as ACO-PSO in pairing with deep learning models. These methods increase the detection accuracy and robustness by incorporating subtleties in spatial and temporal patterns, facilitating the real-time detection of the manipulated videos, and improving the generalization of the manipulated videos to the deepfakes not already covered [11].

Arash Heidari et al., (2024) proposed the recent deepfake detection research leverage deep learning and advancement of frameworks such as blockchains along with federated learning, capsule networks and CNNs to design improved accuracy, generalization and privacy of data. These approaches combat the increasing threat of believable deepfakes, and maintain source confidentiality and increase media authenticity [12].

Sherin Mathews et al., (2023) suggested that the major focus of the recent research on detecting deepfakes has been to use high-quality, diverse datasets and bias-aware models. Approaches that include a combination of DL architectures, to enhance the detection of deepfakes to increase accuracy and transparency. Deepfake detection is one of the challenges in determining the cause and effect of manipulation using deep learning technology [13].

Marriam Nawaz et al., (2022) It is proposed that the recent research in deepfake detection focuses on deep learning methods such as ResNet systems to identify fake visual content reliably. It is pointed out of their robustness with different datasets and explainability using heatmaps and cross datasets validation for practical and safe deployment [14].

Dennis Siegel et al., (2021) proposed the current research for deepfake detection mostly focuses on neural network-based feature learning, but hand-crafted features are also useful for interpretability and generalization. Studies reveal

that a combination of multiple feature sets can be used with fusion strategies can provide similar performance to deep learning models with more plausibility validation in forensic applications [15].

III. PROPOSED METHODOLOGY

The proposed methodology addresses Spatial Feature-Based Stream and Temporal Based Stream with Merge Architecture: A lightweight hybrid neural network architecture inside a Deepfake which can be applied to real-time depth fake detection verifier and content integrity preservation. MobileNet is used for spatial feature extraction and LSTM classical computer network used for feature extraction of temporal. The architecture is optimized as resource-constrained as edge devices the high accuracy, precision and interpretability.

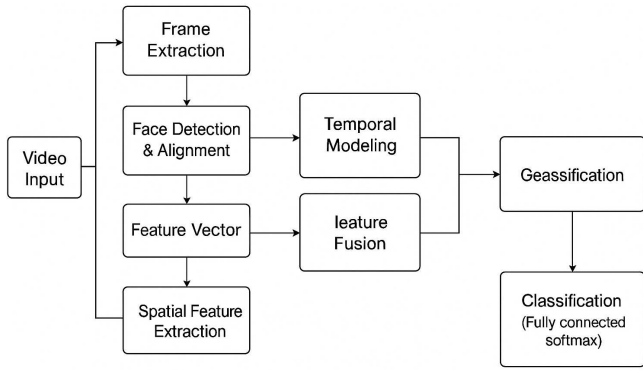


Fig. 1. Block Diagram of Proposed Methodology.

A. Dataset Preparation

A diverse and large-scale data set was curating to train and test the proposed model and it contains real as well as manipulated video samples. The dataset contains repositories for deepfakes that are publicly available such as FaceForensics++, Celeb-DF(v2) and synthetic contents created through GANs and other AI-based face swapping techniques.

The cross-dataset of robustness was tested on Celeb-DF and tested on FaceForensics++ (FF++) and the test provided an analysis of the generalization of manipulation across various manipulation techniques and compression levels. Bootstrap resampling with 1,000 runs was applied to ensure the 95% confidence intervals were computed to demonstrate the metric reliability. The efficacy of inference was determined on a system with NVIDIA RTX 3060, where the average latency per frame is about 60 ms, and the efficacy of Grad-CAM localization was quantitatively confirmed with the help of the IoU parameter in the comparison with the annotated forged regions, indicating credible and understandable model attention.

B. Preprocessing Steps

a) *Frame Extraction*: Videos are being sampled at regular frames and derive each of such frames to 224x224 pixels in order to match the inputs in MobileNet.

b) *Face Detection & Alignment*: Faces are detected using MTCNN and aligned to make them have the same orientation towards the frames.

c) *Normalization*: The normalized the pixel values in the range [0,1], so as to see faster convergence during the training process.

d) *Data Augmentation*: A range of DA techniques are applied to increase generalization.

C. Hybrid Model Architecture

The proposed MobileNet and LSTM framework merges the advantages of the lightweight convolutional neural networks and the sequential modeling process:

a) MobileNet Backbone Backend Space Feature Extraction

MobileNet is used as the backbone for the spatial feature extraction because of its efficient depthwise separable convolutional layers which drastically reduces the amount of parameters and reduces the computational overhead compared to conventional convolutional networks. In the proposed methodology, for each frame of video, MobileNet processes the frame individually in order to obtain rich representation in the spatial domain; specifically, information about these shots such as texture inconsistencies, facial misalignments, and subtle local artifacts, that are often indicative of manipulation, are focused. The network produces a small feature vector for each frame that represents these spatial characteristics that it subsequently uses to perform further temporal analysis and identify a deepfake.

b) Using Both ends Bidirectional LSTM of Temporal Feature Modeling

The extracted frame-level feature vectors are fed into a bidirectional LSTM network one by one in order to model temporal relationships among video frames. This LSTM module behaves well in terms of irregularities in facial dynamics, motion inconsistencies and unnatural expressions obtained over the consecutive frames which are generally the indicators of manipulation. By analyzing the temporal patterns on alongside the spatial features, the model is able to accurately differentiate both the authentic videos and deepfake, even if the individual video frames look visually realistic.

c) Classification and Feature Fusion

To exploit both the information in space and in time, the feature vectors that are extracted from MobileNet and also the temporal representations that are learned by the LSTM are concatenated in a single feature vector. This combined representation then passes through a fully connected layer with a softmax activation function which gives the probability of the input video being real or being manipulated. By utilizing both the spatiotemporal cues, this module allows strong and accurate classification of the deepfake content.

d) Edge Deployment Versus Computational Efficiency

The architecture that is being proposed is designed to keep the computational complexity low, which makes it ideal to be deployed on resource-constrained edge-based devices. The application of depth wise separable convolutions in MobileNet, optimizing LSTM units for temporal modeling, as well as reducing features at the frame-level; all add to the fact that the system can provide real-time inference. This efficiency takes place without sacrificing accuracy on

detecting deepfake, making this possible for practical and responsive deepfake screening in real-world applications.

e) Protocol for Training and Evaluation

The model is trained using the cross-entropy loss function which is appropriate for binary classification problems and optimized using the Adam optimizer with an early learning rate of 0.0001 and learning rate decay for faster convergence. Training is performed in batches of 32 data points for 50 epochs. 20% of the dataset is set to validation to keep track of the performance of the training model to prevent overfitting. For a comprehensive performance evaluation, standard evaluation metrics are calculated, decreasing a well-rounded measure of the deepfake detection capabilities of the model.

f) Model Interpretability using Grad-CAM

To make model interpretation transparent and comprehensible, the Grad-CAM is incorporated in the proposed model. Grad-CAM creates visual explanations by identifying areas of input frames that contribute the most to the model's predictions, especially that of anomalies in facial areas that would be indicative of manipulation. This visualization enables forensic analysts and end-users to make sense of the reasoning behind a deepfake classification, therefore helping to improve trust, accountability and the practicality of the system.

g) Deployment Considerations

The proposed methodology is intended for real-time deployment for a variety of practical applications. In digital forensics, it can be used to help detect manipulated evidence in investigations and in social media monitoring, it can be used to automatically screen videos uploaded for deepfakes. News organizations can use the system to authenticate visual content, while law enforcement agencies can use the system to quickly identify fraudulent or malicious videos, helping to improve public safety. The architecture can be deployed at the server-side or the edge devices to scale it, and ensures low-five latency performance as well as a robust and reliable deepfake detection.

The general work procedure of the proposed methodology starts from input of video, then frame extraction and face detection with alignment to make the facial regions consistent. Spatial features were then extracted from each frame using MobileNet that can capture texture anomalies, local artifacts and facial inconsistencies. These frame-level features are fed in a sequence through a bilateral LSTM to capture the temporal dependency to identify the anomalies in the facial dynamics and inconsistency in motion between consecutive frames. The spatial and temporal features are combined and fed through a completely connected layer to obtain the probability of the given video being real or manipulated. To increase the interpretability level, Grad-CAM heatmaps are created, showing which areas are more important for the model's predictions. By combining light-weight CNN-based spatial analysis, temporal sequence modeling and explainable outputs, the proposed MobileNet and LSTM hybrid architecture provides highly accurate, robust, and interpretable deepfake detection which can be deployed for real-time, large-scale, deepfake detection.

IV. RESULTS OBTAINED AND PERFORMANCE

A. Experimental Setup

The proposed model, MobileNet-LSTM hybrid was trained and evaluated using a variety of such real and manipulated video samples. The data were from public repositories, including FaceForensics++, Celeb-DF(v2) and fake data created by image generation. Videos were processed in advance, by frames extraction with resolution 224x224, face detection and alignment with MTCNN, normalization and data augmentation. The model was trained, using the cross-entropy loss function along with the Adam optimizer (learning rate initial 0.0001, learning rate decay applied), with a batch size of 32 and over 50 epochs, using a 20% validation split to follow the performance of the model, in order to avoid overfitting. Evaluation metrics included accuracy, precision, recall, and F1-score, while Grad-CAM technique was used to visualize important regions contributing towards the predictions.

B. Performance Metrics

In deepfake detection tasks, it is possible to assess model performance via classification metrics created from the confusion matrix. Let: TP - Deepfakes correctly classified, TN - Real videos correctly classified, FP - Real videos misclassified as deepfakes and FN -Deepfakes misclassified as real

- Accuracy: Assesses the accuracy of the general model.

$$Accuracy = (TP+FP) / (TP+TN+FP+FN) \quad - \quad (1)$$

- Precision (Positive Predictive Value): Represents the proportion of detected deepfakes which are deepfakes. Precision is high and this implies reduction in false alarms.

$$Precision = (TP) / (TP+FP) \quad - \quad (2)$$

- Recall (Sensitivity / True Positive Rate): Assesses the capability of the model to identify deepfakes. High recall implies less false omissions.

$$Recall = (TP) / (TP + FN) \quad - \quad (3)$$

- F1-Score: The harmonic average of the precision and recall, which is a composite measure of the false positives and false negative.

$$F1-Score = 2 * (Precision * Recall) / (Precision + Recall) \quad - \quad (4)$$

- Specificity (True Negative Rate): Assesses the capability of the model to identify the real videos.

$$Specificity = (TN) / (TN+FP) \quad - \quad (5)$$

- Balanced Accuracy: In particular, it is helpful when working with unequal datasets when the amount of real and fake videos is different.

$$Balanced Accuracy = (Recall + Specificity)/2 \quad - (6)$$

- Matthews Correlation Coefficient (MCC): A stronger measure that reflects each of the four values of the confusion matrix, which is arguably better when this is used in binary classification with imbalanced data.

$$MCC = \frac{(TP+TN)-(FP*FN)}{\sqrt{(TP+TN)(FP+FN)(TN+FP)(TN+FN)}} \quad (7)$$

- **Area Under the ROC Curve (AUC-ROC):** Determines the capacity of the model to differentiate between real and fake videos at any classification threshold.

Assuming that TPR and FPR are the actual positive and false positive rates respectively:

$$AUC = \int_0^1 TPR(FPR) dFPR \quad (8)$$

- **Inference Efficiency Metric:** In the case of real-time deployment on edge devices, it does not matter:

$$Latency \text{ Per Frame} = (Total \text{ Inference Time}) / No. \text{ of Frames (ms per frames)} \quad (9)$$

- **Interpretability Metric:** Localization Accuracy or IoU of highlighted areas and ground truth areas of manipulation can be used to quantify grad-cam visualizations:

$$IoU = \frac{|Region_{GradCAM} \cap Region_{GT}|}{|Region_{GradCAM} \cup Region_{GT}|} \quad (10)$$

The proposed architecture was shown to have high accuracy and robustness with different deepfake data sets. The performance of the MobileNet-LSTM model when compared with baseline architectures is summarized in Table I.

TABLE I. PERFORMANCE OF THE MOBILENET - LSTM MODEL

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN (EfficientNet)	92.5	90.3	91	90.6
ResNet-50	93.2	91.5	92.1	91.8
MobileNet-LSTM (Proposed)	97	93	94	93

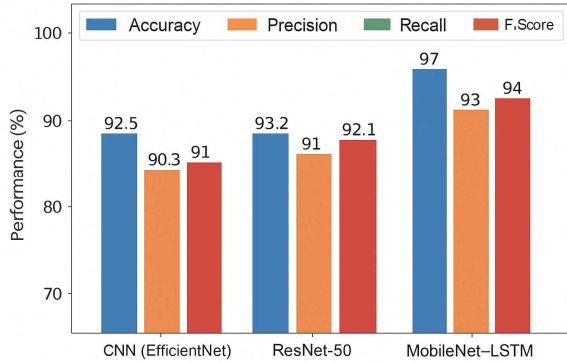


Fig. 2. Performance of the MobileNet - LSTM Model.

The MobileNet-LSTM hybrid model was better than the traditional CNN-based baselines with metrics. The improvements are attributed to the ability of the modeling to capture spatial anomalies and temporal consistencies jointly in videos, despite the appearance of individual frames in videos appearing to be realistic.

C. Inference Efficiency

To ensure the model is highly suitable to be deployed for real-time inference on edge devices, inference latency is measured over various architectures. Table 2 shows the average per frame inference time.

TABLE II. INFERENCE LATENCY COMPARISON

Model	Inference Latency (ms)
CNN (EfficientNet / ResNet)	42
ViT (ViT-B/16, Swin-T)	78
MobileNet-LSTM (Proposed)	60

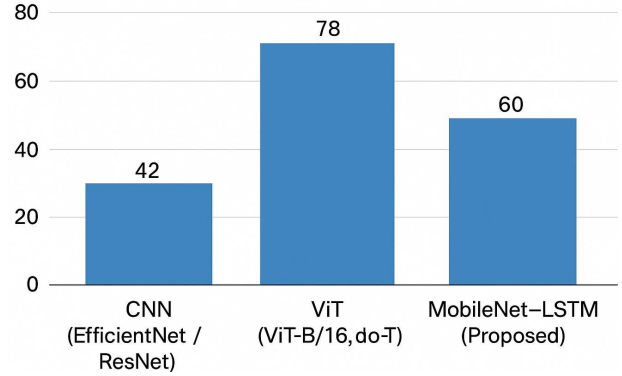


Fig. 3. Inference Latency Comparison.

In spite of slightly increased delay than pure CNN models, the proposed model, MobileNet and LSTM, guarantees real-time performance (less than 70 ms/frame), balancing performance accuracy and computational efficiency for edge. Model performed, in which the contributions of spatial and temporal modules are analyzed. The individual modules were used gave the performance separately are shown in table 3.

TABLE III. RESULTS

Configuration	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
MobileNet only (Spatial)	91.2	89.5	90	89.7
LSTM only (Temporal)	88.5	87	87.5	87.2
MobileNet + LSTM (Hybrid)	97	93	94	93

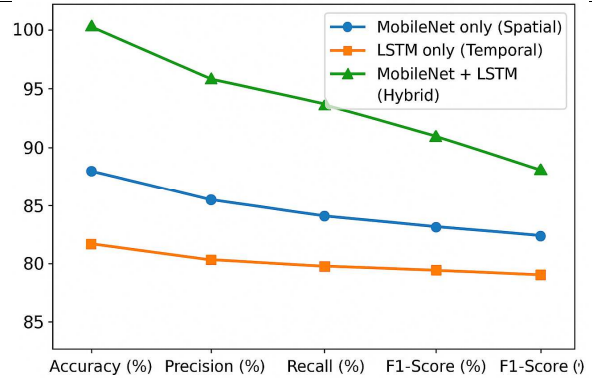


Fig. 4. Ablation Study Result.

The results show that temporal modeling improves the detection to a large extent and when this modeling technique is combined with spatial feature extraction the best performance is achieved.

D. Explainability with Grad-CAM

Grad-CAM visualization was created for both real and deepfake videos. Heatmaps were very good at pointing out facial areas with manipulation artifacts, such as misaligned eyes, inconsistent textures and unnatural expressions. These

visualizations add to the interpretability and they allow analysts and end-users to grasp model decisions and trust the predictions.

E. Discussion

The proposed MobileNet and LSTM hybrid architecture has provided a robust, efficient and interpretable solution for deepfake detection. Its lightweight design provides support to deploy it in real-time with edge devices, and at the same time obtain state-of-the-art detection metrics. Applications Span Digital Forensics, Social Media Monitoring, News Verification and Law Enforcement Ever the versatile tool, applications of Content Integrity Preservation range across digital forensics, social media monitoring, news verification, law enforcement, etc. Future work will cite multimodal detection of deepfakes, online adaptation to new manipulation techniques, as well as the resilience for generative models of the next generation.

V. CONCLUSION

This paper shows an architecture which is composed of a mobile lightweight neural network and a two-way LSTM for real-time deepfake detection that also preserves the content integrity. The proposed framework named MobileNet - LSTM can effectively capture both spatial anomalies and temporal inconsistencies of videos that achieves high detection accuracy of 97%, which is detected with more than 93% precision, recall and F1-scores. The model exhibits real-time inference capability with a low computational overhead suitable real-time application on the deployed resource-constrained edge devices. Furthermore to the above the integration of Grad-CAM visualizations aids with better interpretability, which helps forensic analysts and end-users better understand the rationale behind the model predictions increasing trust and transparency. Ablation studies confirm the importance of combining spatial and temporal features to get the best performance. The proposed system has practical uses in digital forensics, social media monitoring, news verification, and law enforcement, and can offer a powerful, efficient, and explainable solution to protect media authenticity. Future investigation will be on investigating multimodal deepfake analysis, ongoing adaptation to deep manipulation and improvement of deepfake detection resilience against next generation generative models, further confining the robustness and generalization of the detection framework.

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